



Multi-objective optimal design of cycloid speed reducer based on genetic algorithm

Jian Wang^a, Shanming Luo^{a,*}, Deyu Su^b

^a School of Mechanical & Automotive Engineering, Xiamen University of Technology, Xiamen, 361024, PR China

^b Institute of Intelligent Manufacturing of Huazhong University of Science and Technology in Quanzhou, Quanzhou, 362300, PR China

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ABSTRACT

The design of cycloid speed reducer has to satisfy various constraints, e.g. the geometrical, the kinematical and the mechanical, while delivering excellent performance, high efficiency, long life, great load capacity, etc. This invokes the need of an optimal design methodology to achieve these objectives collectively. Taking K-H-V cycloid speed reducers as the research subject, an optimization methodology based on genetic algorithm is presented. The goal of the optimization is to simultaneously minimize volume and maximize efficiency. The volume and the efficiency of the reducer are simultaneously taken as objective functions. The design variables including the short width coefficient, the diameter of the pin, the width of the cycloid gear, etc. are defined. The constraint conditions are established for multi-optimizations. In order to elaborate the proposed optimization methodology, the XW3-type cycloid speed reducer is chosen and utilized as an example for optimization. A comparison study on the single optimization is also carried out in this work. The research results are satisfactory and can help designers to employ for minimum material and cost by fulfilling high efficiency and other performance requirements.

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1. Introduction

Cycloid speed reducers have found wide applications in the automation field as industry robots, machine tools, automatic machinery and other aspects owing to their excellent characteristics, e.g. high transmission accuracy, large reduction ratio, great load capacity, high efficiency, low backlash, long service life, high shock load capacity, etc. Therefore, the research and development of the cycloid speed reducers have been investigated by many researchers, including operating principle, conditions for non-undercutting manufacturing, force analysis and efficiency, and development of new mechanisms [1–6].

In general, the cycloid speed reducers can be classified into four types according to the lobe profile of the cycloid plate gear and the motion of the roller gear. They are the stationary ring gear type epicycloid reducers, the rotating ring gear type epicycloid reducers, the stationary ring gear type hypocycloid reducers and the rotating ring gear type hypocycloid reducers. But no matter what type it is, a cycloid speed reducer contains the transmission part, the input part and output part, etc. These components are interrelated and interact on each other, resulting that the running status and the output characteristics of the cycloid speed

* Corresponding author at: School of Mechanical & Automotive Engineering, Xiamen University of Technology, No. 600, Ligong Road, Jimei District, Xiamen City, Fujian Province, 361024, China.

E-mail address: s.luo@hotmail.com (S. Luo).

reducers will be influenced by one of the parameters of these components. For instance, the impact of a certain choice of gear width may yield a minimum mass gearing, but the selection of this gearing may cascade through subsequent steps of the design process (sizing of shafts, further stages, bearings, etc.) to ultimately lead to a heavier reducer. In addition, in the design of the cycloid speed reducer, designers always consider certain objectives such as efficiency, weight, load capacity, contact strength depending on the requirements. There also exist interactions and constraints between them, e.g., if module is calculated based on the objective of bending strength, the same module is substituted to calculate the objective of surface durability, it is accepted if it is within the strength limit of surface durability; otherwise it is changed accordingly. So optimization methods are required to determine design variables which simultaneously satisfy the given conditions. Moreover, increasing demand for compact, efficient, and reliable cycloid speed reducers forces the designer to use optimal design methodology.

There have been a number of studies attempting to optimize gears or gear reducers with the aid of computers. Thompson [7] presented a generalized approach to the minimum volume design of spur gear units based upon a traditional nonlinear programming. The methodology was applied to the design of two-stage and three-stage spur gear reduction units, subject to identical loading conditions and design criteria. Senthil Kumar [8] carried out the optimization of the asymmetric spur gear drive using an iterative procedure on the calculated maximum fillet stresses through FEM for different rack cutter shifts. The optimum values of rack cutter shifts were suggested for the given center distance and the speed ratio of an asymmetric gear drive. Tudose [9] discussed a two-stage helical gear transmission design problem (complete with the sizing and selection of shafts, bearings, housing, etc.) using a two-phase evolutionary algorithm in a formulation that can be extended to include additional stages or different layouts. Savsani [10] presented two advanced optimization algorithms known as particle swarm optimization (PSO) and simulated annealing (SA) to find the optimal combination of design parameters for minimum weight of a spur gear train. Buiga [11] carried out the complete automated optimal design with genetic algorithms of a two-stage helical coaxial speed reducer. The objective function (i.e. the mass of the entire speed reducer) was described by a set of 17 mixed design variables (i.e. integer, discrete and real) and also was subjected to 76 highly non-linear constraints. The results showed that the proposed genetic algorithm can offer better design solutions as compared with the results obtained by using the traditional design method. Tamboli [12] analyzed the helical gear pair for minimum volume using particle swarm optimization technique, since the most power transmission systems require low weight energy efficient and cost effective system elements. Wan [13] carried out the optimization of cycloid gears. The constraints were the cycloid gear can be mounted successfully at any place and the curve was without interference. The optimum objective was that the strength and stiffness of the pin gear housing weaken was least serves. Wang [14] proposed a new method to determine the dynamic model of a practical gear system by the identification of the relation between the rotational movement of gears and the acoustical noise signal measured on the gear tester. The optimized gear tooth profile modification was derived based on the identified model. The tested result showed that the profile modification can greatly reduce gear noise. Li [15] analyzed the realization of the generalized reduced gradient method and its application in a single-stage cylindrical gear reducer design. The experimental results showed that the proposed method has fewer iterations and higher precision. He [16] carried out an optimal design method to reduce vibration and noise of double crank ring-plate-type pin-cycloid planetary drive. Zhang [17] established an optimization model, which included discrete variables (module), integer variables (teeth number) and continuous variables (tooth width, spiral angle, etc.), and proposed a new optimization solution algorithm to obtain the discrete solution of module and teeth number.

An engineering design optimization problem is generally composed of multiple objectives, associated with a constrained parameter space, thereby getting the name as (constraint) multi-objective optimization. Formally, the multi-objective optimization refers to the solution of problems with two or more objective functions, which are normally in conflict with each other. Nowadays, only a few references can be found on multi-objective optimal design of reducers. Gao [18] carried out the multi-objective optimizations of the harmonic reducer. On the basis of the optimization results, a new type short cylindrical cup-shaped harmonic reducer was designed and manufactured. Wang [19] established a multi-objective optimization model of optimizing contact force and reducer volume in order to reduce the gearbox volume and the impact of running while improving bearing capacity. Wang [20] presented a multi-objective optimization design method for helical gear reducer based on simulated annealing algorithms. The simulation experiment indicated that the optimal result was superior to that of traditional design scheme. Zhang [21] constructed an optimization model for planetary gear reducer in shield tunneling machine with four objective functions, that was to make its volume smallest, its efficiency highest and the reliability of its contacting and bending strength highest, which were subject to the conditions such as teeth, modification coefficient, interference, strength, equal strength and equal life. In order to reduce the gearbox volume and the impact of running while improving bearing capacity, Wang [22] established a multi-objective optimization model, in which five reducer dimensions were selected as the design variables to construct eight nonlinear constraint equations.

These studies referenced above have been instrumented in order to highlight two important aspects regarding the topic of this work. Firstly, it is obvious that most of the references cited above applied the optimization techniques for individual components (only gears or shafts) or intermediate assemblies rather than to the entire speed reducer (considering the multiple connections between its subsystems, i.e. the gearings, the shafts-subassembly, and the housing). Secondly, only a few references focus on multi-objective optimal design of reducers, especially on cycloid speed reducers. The aim of this work is to establish a multi-objective optimization model of optimizing volume and efficiency of K-H-V cycloid speed reducers. A comparison study on single-objective optimization of cycloid speed reducers, whose objective function is volume and efficiency, respectively, is also carried out in this work.

The remainder of this paper is organized as follows. Section 2 calculates the efficiency and volume of the K-H-V cycloid speed reducer. In Section 3, a mathematical model of multi-objective optimization is established. Section 4 presents an example of the multi-objective optimization. A comparison study of the single objective option design for the same cycloid speed reducer is also carried out in this section. In Section 5, the obtained results of the multi-objective optimization design are compared with that of the single objective optimization design. Finally, a conclusive summary of this study is given in Section 6.

2. Performance of the K-H-V cycloid speed reducer

Having advantages of large transmitting ratio, simple construction, compactness and smooth running, the planetary gearing with small tooth number difference has been widely used. It is also known as the K-H-V planetary gearing. Furthermore, most cycloid speed reducers which produced by domestic factories are K-H-V cycloid speed reducers at present. Therefore, the K-H-V cycloid speed reducer is chosen as the research subject in this paper.

The K-H-V cycloid speed reducer consists of cycloid gear *g*, pinwheel *b*, carrier *H*, output part *W*, etc. As shown in Fig. 1, this type of reducer contains one cycloid gear, 44 pin wheels (also called pin gear), 8 pin sleeves, 8 pins and dual eccentric sleeves. Since the external spur gear uses the cycloid curve as the tooth profile, here it is simply called the cycloid gear. On the cycloid gear, there are 8 holes used to insert the 8 pin sleeves so, these holes are called pin holes in this paper.

In Fig. 1, the pin wheel is used as an internal spur gear to engage with the cycloid gear. Usually the number of pins is one greater than the number of teeth of the cycloid gear. The 8 pin sleeves are mounted on the 8 pins. The pin sleeves are used as the sliding bearings between the pins and the pin holes. The dual eccentric sleeve is inserted into the center hole on the cycloid gear. It is used to produce eccentric movement of the cycloid gear.

2.1. Efficiency of the cycloid speed reducer

Supposing that η_k denotes the meshing efficiency between the pin wheel and the cycloid gear, η_{sx} presents the efficiency of the output part, η_{zx} is the efficiency of the pivoted arm bearing, and η_{gx} is the efficiency of the rolling bearing, the efficiency of the K-H-V cycloid speed reducer, whose pin wheel is fixed and pivoted arm is active, can be expressed as

$$\eta = \eta_k \eta_{zx} \eta_{gx}^2 \eta_{sx} \quad (1)$$

2.1.1. Meshing efficiency between the pin wheel and the cycloid gear

According to Ref. [23], the meshing efficiency η_k between the pin wheel and the cycloid gear can be shown as

$$\eta_k = \frac{\eta_{nx}}{1 + z_c(1 - \eta_{nx})} \quad (2)$$

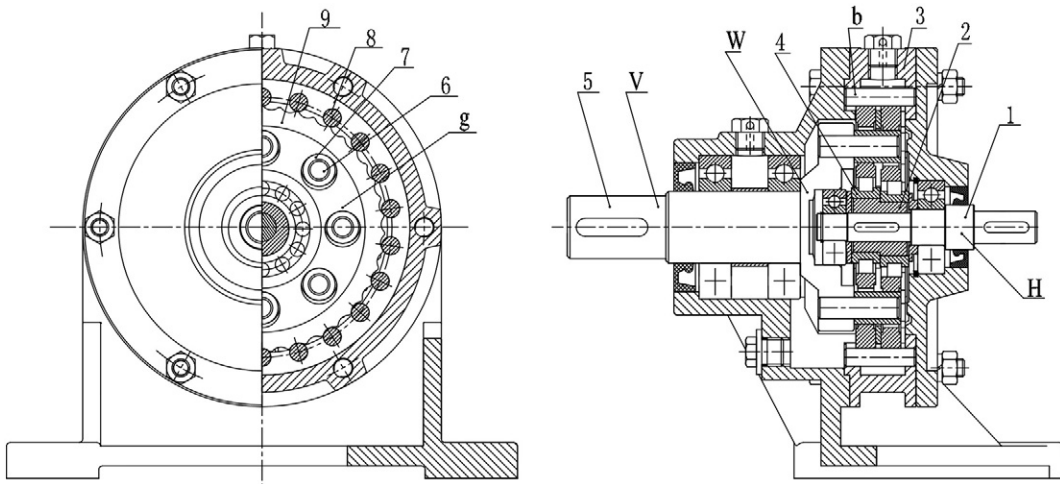


Fig. 1. The structure of cycloid drive (1. Input shaft, 2. Dual eccentric sleeves, 3. Pin gear housing, 4. Pivoted arm bearing, 5. Output shaft, 6. Pin, 7. Pin sleeve, 8. Pin wheel, 9. Cycloid gear).

where z_c represents the tooth number of the cycloid gear, η_{nx} denotes the efficiency of the transforming mechanism of the cycloid speed reducer, which can be expressed as

$$\eta_{nx} = 1 - (D_p - d_{rp}) \frac{4\mu}{K_1 z_c D_p \pi} \quad (3)$$

Hence, Eq. (2) can be rewritten as

$$\eta_x = \frac{1 - (D_p - d_{rp}) \frac{4\mu}{K_1 z_c D_p \pi}}{1 + (D_p - d_{rp}) \frac{4\mu}{K_1 D_p \pi}} \quad (4)$$

where D_{rp} and D_p denote the diameters of the pin wheel and the pin wheel central circle, respectively, K_1 is the short width coefficient, μ is the friction coefficient between the pin wheel and the cycloid gear, which is usually equal to 0.05–0.1.

2.1.2. Efficiency of output part

According to Ref. [23], the efficiency of the output part can be deduced to

$$\eta_{sx} = 1 - \frac{4f_w K_1 d_{sw} D_p}{\pi D_w (d_{sw} + 2\Delta_2)} \quad (5)$$

where D_{sw} and D_w denote the diameters of the pin and the pin central circle, respectively, Δ_2 is wall thickness of the pin sleeve, f_w is the friction coefficient of the output part, which is usually equal to 0.008–0.08.

2.1.3. Efficiency of cycloid speed reducer

According to Eq. (1), the total efficiency of the reducer is not only influenced by the meshing efficiency between the pin wheel and the cycloid gear η_x , the efficiency of the output part η_{sx} , but also influenced by the efficiency of the pivoted arm bearing η_{zx} and the efficiency of the rolling bearing η_{gx} . The efficiency of the rolling bearing is usually equal to 0.99–0.999. We assign the value of $\eta_{gx} = 0.995$. The efficiency of the pivoted arm bearing is smaller than that of the rolling bearing. We assign the value of $\eta_{zx} = 0.99$. Substituting Eqs. (4) and (5) into Eq. (1), the total efficiency of the reducer can be finally expressed as

$$\eta = \eta_x \eta_{zx} \eta_{gx}^2 \eta_{sx} = \frac{1 - (D_p - d_{rp}) \frac{4\mu}{K_1 z_c D_p \pi}}{1 + (D_p - d_{rp}) \frac{4\mu}{K_1 D_p \pi}} \left(1 - \frac{4f_w K_1 d_{sw} D_p}{\pi D_w (d_{sw} + 2\Delta_2)} \right) \eta_{zx} \eta_{gx}^2 \quad (6)$$

According to Eq. (6), the efficiency of the cycloid speed reducer is related to the diameter of the pin wheel d_{rp} , the diameter of the pin wheel central circle D_p , the diameter of the pin d_{sw} , the diameter of the pin central circle D_w , the friction coefficient between the pin wheel and the cycloid gear μ , tooth number of the cycloid gear z_c (transmission ratio i), and short width coefficient K_1 . With the changes of the above parameters, the tooth profile of the cycloid gear will be change. Thus the efficiency has much concern with tooth profile of the cycloid gear.

2.2. Volume of the cycloid speed reducer

According to Fig. 1, the outer diameter of the cycloid gear impacts the radial dimension, and the width of the cycloid gear impacts the axial dimension. Therefore, the volume of the cycloid gear impacts that of the cycloid speed reducer. The volume of the cycloid gear equals to the volume of disk within the dedendum circle added to that of the gear teeth, minus that of the pivoted arm bearing and the pin hole. It can be expressed as [24]

$$V = \frac{1}{4} \pi B \left[\left(D_p - K_1 \frac{D_p}{z_p} - d_{rp} \right)^2 - \left(d_{sw} + 2\Delta_2 + K_1 \frac{D_p}{z_p} \right)^2 z_w - D^2 \right] + K_1 \frac{D_p}{z_p} z_c B \quad (7)$$

where D represents the diameter of the cycloid gear center hole, B represents the width of the cycloid gear.

According to Eq. (7), the volume of the cycloid speed reducer is related to the diameter of the pin wheel d_{rp} , the diameter of the pin wheel central circle D_p , the diameter of the pin d_{sw} , the diameter of the cycloid gear center hole D , the width of the cycloid gear B , tooth number of the cycloid gear z_c (transmission ratio i), short width coefficient K_1 .

3. Mathematical model of multi-objective optimization

Minimizing the volume and maximizing the efficiency of the cycloid speed reducer simultaneously are regarded here as a mathematical optimization problem. The formulation of this problem requires that the design variables, objective functions, and constraints (for defining feasible design parameter space) be defined. We discuss in brief, these design parameters, objective functions, and constraints of the present work in following sections.

3.1. Definition of design variables

According to Eqs. (6) and (7), design variables can be defined. They are the diameter of the pin wheel central circle (D_p), the diameter of the pin wheel (d_{rp}), the diameter of the pin (d_{sw}), short width coefficient (K_1), the diameter of the pin central circle (D_w), the diameter of the cycloid gear center hole (D), the width of the cycloid gear (B). Thus, the design parameter vector can be written as

$$X = [D_p, d_{rp}, B, D, K_1, D_w, d_{sw}]^T \quad (8)$$

3.2. Establishment of objective functions

The efficiency, which is one important criterion to evaluate a speed reducer, should be increased to the best of our abilities during the design process. Furthermore, volume is another important criterion for a speed reducer. Therefore, in this paper, the efficiency η and the volume V of cycloid speed reducers are respectively viewed as objective functions for multi-objective optimization.

Representing one of the objective functions in the form as below

$$\min f_1 = 1 - \frac{1 - (D_p - d_{rp}) \frac{4\mu}{K_1 z_c D_p \pi}}{1 + (D_p - d_{rp}) \frac{4\mu}{K_1 D_p \pi}} \left(1 - \frac{4f_w K_1 d_{sw} D_p}{\pi D_w (d_{sw} + 2\Delta_2)} \right) \eta_{zx} \eta_{gx}^2 \quad (9)$$

Another objective function is defined by

$$\min f_2(x) = \frac{1}{4} \pi B \left[\left(D_p - K_1 \frac{D_p}{z_p} - d_{rp} \right)^2 - \left(d_{sw} + 2\Delta_2 + K_1 \frac{D_p}{z_p} \right)^2 z_w - D^2 \right] + K_1 \frac{D_p}{z_p} z_c B. \quad (10)$$

3.3. Constraint conditions

For satisfying the actual requirement of the project, the selection of short width coefficient and pin-diameter coefficient should be within the recommended range, the conditions including top cutting or sharp angle will be discussed in design of the cycloid gear, the constraints including the contact and bending strength of reducer components, life of the pivoted arm bearing etc. will also be established. These constraint conditions are discussed in more detail in the following subsections.

3.3.1. The condition for avoiding top cutting or sharp angle

As to the convex tooth profile of cycloid gear, the smallest curvature radius $|\rho_0|_{\min}$ can be expressed as [25]

$$|\rho_0|_{\min} = \begin{cases} \frac{D_p}{2} \sqrt{\frac{27(1-K_1^2)(z_p-1)}{(z_p+1)^3}} & \frac{z_p-2}{2z_p-1} < K_1 < 1 \\ \frac{D_p(1-K_1)^2}{2(z_p K_1 + 1)} & K_1 \leq \frac{z_p-2}{2z_p-1} \end{cases} \quad (11)$$

Table 1
Suggested value of K_1 .

z_c	≤ 11	13–23	25–59	61–87
K_1	0.42–0.55	0.48–0.74	0.65–0.9	0.75–0.9

Therefore, the condition for avoiding top cutting or sharp angle can be derived as follows:

$$\begin{aligned} \frac{d_{rp}}{2} - \frac{D_p}{2} \sqrt{\frac{27(1-K_1^2)(z_p-1)}{(z_p+1)^3}} < 0 \quad \frac{z_c-1}{2z_c+1} < K_1 < 1 \\ \frac{d_{rp}}{2} - \frac{D_p(1-K_1)^2}{2z_pK_1+1} < 0 \quad K_1 \leq \frac{z_c-1}{2z_c+1} \end{aligned} \quad (12)$$

3.3.2. The constraint condition of short width coefficient

The short width coefficient is an important design variable of the cycloid speed reducer, which directly determines the property of the reducer. According to Ref. [25], reasonable values for short width coefficient are suggested in Table 1.

Supposing that the constraint range of the short width coefficient is $[c_1, c_2]$, the constraint equation can be defined by

$$c_1 - K_1 \leq 0 \quad (13)$$

$$K_1 - c_2 \leq 0 \quad (14)$$

3.3.3. The constraint condition of pin-diameter coefficient

The selection value of the radius of pin wheel d_{rp} should not only meet the requirement of collision avoidance for pin wheels, but also guarantee enough strength for pin wheel and pin wheel housing. The relationship between the radius of pin wheel r_{rp} and pin-diameter coefficient K_2 can be expressed as

$$K_2 = \frac{D_p}{d_{rp}} \sin \frac{\pi}{z_p}. \quad (15)$$

According to Ref. [25], the suggested values of pin-diameter coefficient K_2 are listed in Table 2. Similarly, supposing that the selection range of pin-diameter coefficient is $[d_1, d_2]$, the constraint equation can be defined by

$$d_1 - \frac{D_p}{d_{rp}} \sin \frac{\pi}{z_p} \leq 0 \quad (16)$$

$$\frac{D_p}{d_{rp}} \sin \frac{\pi}{z_p} - d_2 \leq 0. \quad (17)$$

3.3.4. The contact strength of the cycloid gear

According to Ref. [25], using Hertz theory, the contact stress between the pin gear and the cycloid gear can be expressed as

$$\sigma_H = 0.418 \sqrt{\frac{E_e F_i}{B \rho_{ei}}} \leq \sigma_{HP} \quad (18)$$

where E_e is the equivalent elastic modulus, ρ_{ei} is the equivalent curvature radius, F_i is the meshing force in a certain position between the pin gear and the cycloid gear, σ_{HP} is the allowable contact stress.

Supposing that σ_{Hmax} represents the maximum contact stress, according to Eq. (18), it can be defined by

$$\sigma_{Hmax} = 0.418 \sqrt{\frac{E_e F_{max}}{B \rho_{emin}}} \quad (19)$$

where ρ_{emin} represents the minimum equivalent curvature radius, F_{max} denotes the maximum meshing force between the pin gear and the cycloid gear. In order to shorten the running time, F_{max} can be calculated using the following equation [25].

$$F_{max} = \frac{4.4M}{K_1 z_c D_p} \quad (20)$$

where M is the torque of output shaft.

Hence, Eq. (19) can be rewritten as

$$\sigma_{H\max} = 0.418 \sqrt{\frac{E_e}{BK_1 z_c D_p} \frac{4.4M}{\rho_{\min}}} \quad (21)$$

The constraint equation can be finally defined by

$$0.418 \sqrt{\frac{E_e}{BK_1 z_c D_p} \frac{4.4M}{\rho_{\min}}} - \sigma_{HP} \leq 0. \quad (22)$$

3.3.5. The bending strength of the pin wheel

During the meshing period, the meshing force between the cycloid gear and the pin wheel will cause bending deformation of the pin wheel. The excessive bending deformation will certainly lead to bad contact, which will make it easier for veneering failure either between the cycloid gear and the pin wheel. Therefore, it is necessary to check the bending strength of the pin wheel.

Generally, when $D_p < 390$ mm, pin wheel with two pivots is used. Otherwise, pin wheel with three pivots is used. According to Ref. [25], as for pin wheel with two pivots, the bending stress of the pin wheel can be expressed as

$$\sigma_F = \frac{1.41 F_{\max} L}{d_{rp}^2} \leq \sigma_{FP}. \quad (23)$$

As for pin wheel with three pivots, the bending stress can be expressed as follows.

$$\sigma_F = \frac{0.48 F_{\max} L}{d_{rp}^2} \leq \sigma_{FP}. \quad (24)$$

Hence, the constraint equation can be finally defined by

$$\begin{cases} \frac{1.41 \times 4.4 \times 9550PL}{K_1 D_p n d_{rp}^2} - \sigma_{FP} \leq 0 & D_p < 390 \\ \frac{0.48 \times 4.4 \times 9550PL}{K_1 D_p n d_{rp}^2} - \sigma_{FP} \leq 0 & D_p \geq 390 \end{cases} \quad (25)$$

3.3.6. The contact strength between the pin and the pin hole

According to Ref. [26], the contact strength between the pin and the pin hole can be expressed as

$$\sigma_{ZH} = 300 \sqrt{\frac{K_1 M D_p}{z_w D_w B \left((r_{sw} + \Delta_2)^2 z_p + \frac{1}{2} K_1 D_p (r_{sw} + \Delta_2) \right)}} \leq \sigma_{ZHP}. \quad (26)$$

The constraint equation can be defined by

$$300 \sqrt{\frac{K_1 M D_p}{z_w D_w B \left(\left(\frac{d_{sw}}{2} + \Delta_2 \right)^2 z_p + \frac{1}{2} K_1 D_p \left(\frac{d_{sw}}{2} + \Delta_2 \right) \right)}} - \sigma_{ZHP} \leq 0. \quad (27)$$

Table 2
Suggested value of K_2 .

z_p	<12	12–24	24–36	36–60	60–88
K_2	3.85–2.85	2.8–2.0	0–1.25	1.6–1.0	1.5–0.99

Table 3Suggested values of D_p .

types	0	1	2	3	4	5	6	7
D_p/mm	75–94	95–105	106–120	140–155	165–185	210–230	250–275	280–300

3.3.7. The bending strength of the pin

The bending strength of the pin can be expressed as [26]

$$\sigma_w = \frac{K_w Q_{\max}(1.5B + \Delta_c)}{0.1d_{sw}^3} \leq \sigma_{BP} \quad (28)$$

where Δ_c is the thickness of the space ring, K_w is the influence coefficient of manufacturing error and installation error on load of pin, σ_{BP} is the allowable bending stress. For simple applications, the suggested value of K_w is 1.4.

The constraint equation can be defined by

$$\frac{4.4K_w M(1.5B + \Delta_c)}{0.1z_w D_w d_{sw}^3} - \sigma_{BP} \leq 0 \quad (29)$$

3.3.8. The constraint condition of the diameter of the pin central circle

The suggested values of the diameter of the pin central circle are listed in Table 3, corresponding to different type of cycloid speed reducers.

Supposing that the constraint range of the diameter of pin central circle is $[e_1, e_2]$, the constraint equation can be expressed as

$$e_1 - D_p \leq 0 \quad (30)$$

$$D_p - e_2 \leq 0. \quad (31)$$

3.3.9. The constraint condition of the maximum diameter of the pin hole

In order to ensure the cycloid gear has strong strength, there must have a certain thickness Δ_1 , which is usually no less than $0.03D_p$, between cycloid gear center hole and two pin holes. According to Ref. [25], the constraint condition can be expressed as

$$0.06D_p - D_w + d_w + D \leq 0 \quad (32)$$

$$0.03D_p - D_w \sin \frac{\pi}{z_w} + d_w \leq 0 \quad (33)$$

where d_w denotes the diameter of pin hole, and it can be defined by

$$d_w = d_{sw} + 2\Delta_2 + 2a + \Delta \quad (34)$$

where a denotes the centre distance, Δ represents the clearance between the pin hole and the pin sleeve, which is usually equal to 0.15 mm.

3.3.10. Width of cycloid gear

The width of the cycloid gear is usually equal to $0.05D_p - 0.1D_p$. The constraint equation can thus be defined by

$$0.05D_p - B \leq 0 \quad (35)$$

$$B - 0.1D_p \leq 0. \quad (36)$$

3.3.11. Life of pivoted arm bearing

According to Ref. [25], the life of the pivoted arm bearing L_h can be expressed as

$$L_h = \frac{10^6}{60n_1} \left(\frac{C}{p} \right)^{\frac{10}{3}} \quad (37)$$

where C represents the rated dynamic load of the bearing, p denotes the dynamic load of the bearing, R is the nominal radial load of the bearing, n_1 is the rotate speed of the pivoted arm bearing. Moreover, the following relationships exist:

$$\begin{aligned} C &= 9.8 \times 6.4 \times (0.1R_p)^{1.85} (2.065 + 0.91R_p)^{0.75} \\ R &= \frac{0.825Mz_p}{K_1 D_p z_c} \\ p &= 1.25R \\ n_1 &= \frac{nz_p}{z_p - 1} \end{aligned}$$

where n represents the rotate speed of the input shaft.

The life of the pivoted arm bearing L_h is usually equal to 5000 h. The constraint equation can thus be expressed as

$$5000 - \frac{10^6}{60n_1} \left(\frac{C}{p} \right)^{\frac{10}{3}} \leq 0. \quad (38)$$

3.4. Mathematical model

The vector of design variables is shown as

$$X = [x_1, x_2, x_3, x_4, x_5, x_6, x_7]^T = [D_p, d_{rp}, B, D, K_1, D_w, d_{sw}]^T.$$

The objective function is defined by

$$\begin{cases} \min f_1(x) = 1 - \frac{1 - (x_1 - x_2) \frac{4\mu}{x_5 z_c x_1 \pi}}{1 + (x_1 - x_2) \frac{4\mu}{x_5 x_1 \pi}} \left(1 - \frac{4f_w x_5 x_7 x_1}{\pi x_6 (x_7 + 2\Delta)} \right) \eta_{zx} \eta_{gx}^2 \\ \min f_2(x) = \frac{1}{4} \pi x_3 \left[\left(x_1 - x_5 \frac{x_1}{z_p} - x_2 \right)^2 - \left(x_7 + 2\Delta_2 + x_5 \frac{x_1}{z_p} \right)^2 z_w - x_4^2 \right] + x_5 \frac{x_1}{z_p} z_c x_3 \end{cases}$$

Table 4

Initial values of design variables.

Design variables	Symbol	Unit	Interval	Initial value
Diameter of pin wheel central circle	D_p	mm	$140 \leq D_p \leq 155$	144
Diameter of pin wheel	d_{rp}	mm	$7 \leq d_{rp} \leq 10.4$	10
Width of cycloid gear	B	mm	$7 \leq B \leq 12$	11
Diameter of cycloid gear center hole	D	mm	$50 \leq D \leq 55$	53.5
Short width coefficient	K_1	–	$0.65 \leq K_1 \leq 0.9$	0.6069
Diameter of pin central circle	D_w	mm	$88 \leq D_w \leq 104$	90
Diameter of pin	d_{sw}	mm	$11 \leq d_{sw} \leq 14$	12

The constraints can be formed as follows. A feasible design must satisfy all constraints that all $y_i(x)$ function values must be less than or equal to zero. These constraints are inequality constraints and there is no equality constraint in the problem considered.

$$\begin{aligned}
 y_1(x) &= \begin{cases} \frac{x_2}{2} - \frac{x_1}{2} \sqrt{\frac{27(1-x_5^2)(z_p-1)}{(z_p+1)^3}} < 0 & 1 > x_5 > \frac{z_c-1}{2z_c+1} \\ \frac{x_2}{2} - \frac{x_1}{2} \frac{(1-x_5)^2}{z_p x_5 + 1} < 0 & x_5 \leq \frac{z_c-1}{2z_c+1} \end{cases} \\
 y_2(x) &= c_1 - x_5 \leq 0 \\
 y_3(x) &= x_5 - c_2 \leq 0 \\
 y_4(x) &= d_1 - \frac{x_1}{x_2} \sin \frac{\pi}{z_p} \leq 0 \\
 y_5(x) &= \frac{x_1}{x_2} \sin \frac{\pi}{z_p} - d_2 \leq 0 \\
 y_6(x) &= 0.418 \sqrt{\frac{E_e}{x_3 x_5 z_c x_1} \frac{4.4M}{\rho_{emin}}} - \sigma_{HP} \leq 0 \\
 y_7(x) &= \begin{cases} \frac{1.41 \times 4.4 \times 9550PL}{x_5 x_1 n x_2^2} - \sigma_{FP} \leq 0, x_1 < 390 \\ \frac{0.48 \times 4.4 \times 9550PL}{x_5 x_1 n x_2^2} - \sigma_{FP} \leq 0, x_1 \geq 390 \end{cases} \\
 y_8(x) &= 300 \sqrt{\frac{x_5 M x_1}{z_w x_6 x_3 \left(\left(\frac{x_7}{2} + \Delta_2 \right)^2 z_p + \frac{1}{2} x_5 x_1 \left(\frac{x_7}{2} + \Delta_2 \right) \right)}} - \sigma_{ZHP} \leq 0 \\
 y_9(x) &= \frac{4.4K_w M (1.5x_3 + \Delta_c)}{0.1z_w x_6 x_7^3} - \sigma_{BP} \leq 0 \\
 y_{10}(x) &= e_1 - x_1 \leq 0 \\
 y_{11}(x) &= x_1 - e_2 \leq 0 \\
 y_{12}(x) &= 0.06x_1 - x_6 + x_4 + \left(x_7 + 2\Delta_2 + \frac{x_5 x_1}{z_p} + 0.15 \right) \leq 0 \\
 y_{13}(x) &= 0.03x_1 - x_6 \sin \frac{\pi}{z_w} + \left(x_7 + 2\Delta_2 + \frac{x_5 x_1}{z_p} + 0.15 \right) \leq 0 \\
 y_{14}(x) &= 0.05x_1 - x_3 \leq 0 \\
 y_{15}(x) &= x_3 - 0.1x_1 \leq 0 \\
 y_{16}(x) &= 5000 - \frac{10^6}{60n_1} \left(\frac{C}{p} \right)^{\frac{10}{3}} \leq 0
 \end{aligned}$$

4. Optimization example

According to the characteristic of the mathematical model established above, the mathematical model is attributed to a non-linear programming problem. As such, it can be solved by the multi-objective optimization function in the MATLAB software. In order to demonstrate the feasibility of the computer program for multi-objective option design, an example is given. A comparison of the computer program of the single objective option design for the same cycloid speed reducer is also carried out in this work.

The XW3-type cycloid speed reducer was optimized in the example. The value of the transmission ratio i was 43. GCr15 material was selected for cycloid gear, pin wheel, pin and pin sleeve of cycloid speed reducer. The input speed n was required, and its value was 1440 r/min. The input power P was set at 0.75 Kw. The values of the contact and bending stresses were $\sigma_{HP} = 1000$ MPa, $\sigma_{FP} = 150$ MPa and $\sigma_{BP} = 200$ MPa, respectively. The initial values of design variables were listed in Table 4.

4.1. Multi-objective optimization

In this paper, a methodology based on Genetic Algorithms is used to automate the search for the best values. Genetic Algorithms is a “survival of the fittest” optimization algorithm based on evolutionary theory. The “genetic algorithm” concept was first proposed by a student of Professor John Henry Holland in the University of Michigan in 1967. To implement Genetic Algorithms, a genetic representation of feasible solution, namely individual, and the fitness function to evaluate the candidate solutions are required. As shown in Fig. 2, the algorithm begins by establishing an initial population of individuals. Then, each solution is evaluated to measure its quality by assigning a value equivalent to the performance according to the fitness function. The initial

population should be improved through several iterations. In each iteration, three operators, selection, crossover, and mutation, are performed to generate a new population.

In this subsection, the following Genetic Algorithms parameters were used: (a) Population size of 300, (b) Mutation rate of 0.01, (c) Crossover rate of 0.8. The algorithm is stopped if maximum number of iterations is exceeded, here 400, or if no change on the global minimum after occurs 30% of the maximum number of iterations.

According to Section 3, the transmission ratio should be discrete value. Using MATLAB software, the optimization results can be obtained. The comparison parameters before and after optimization are shown in Table 5. The optimization results must be rounded to satisfy practical situations. The rounded values are also listed in Table 5. According to the results, the following conclusions can be made:

- (i) Substituting the optimization variables into above constraint conditions, all of them can satisfy the constraint conditions.
- (ii) The value of the efficiency is increased by about 3.79% compared to the value before optimization, and the value of the volume is 32.51% less than that of the value before optimization.
- (iii) In the results, the values the diameter of the pin wheel central circle and the width of the cycloid gear are smaller than that of the value before optimization, respectively. The results demonstrated that axial and radial sizes of reducers are both reduced, resulting that the structure is more compact. On the contrary, the diameter of the pin central circle is larger than that of the value before optimization, which is beneficial to decrease the contact stress between the pin and the cycloid gear, to increase the efficiency of the reducer.

4.2. Optimization based on minimum volume

In this subsection, a comparison study of the optimization based on minimum volume is carried out. The diameter of the pin central circle is a constant value, which is chosen as 90 mm, while the other design variables keep the same as Table 5. Genetic algorithms parameters used were the same as in the case of the multi-objective optimization. In order to end algorithm, all chromosomes in the population were converged to the best chromosome. In the optimization process, the variation of volume is shown in Fig. 3. Table 6 shows the best results for the optimization of the objective.

From above discussion, the following conclusions to the optimization based on minimum volume can be drawn: (i) All of optimization variables can satisfy the constraint conditions. (ii) The value of the volume is decreased by about 36.08%, and the efficiency is additionally increased by about 3.79% compared to the value before optimization. (iii) The values the diameter of the pin wheel central circle and the width of the cycloid gear are smaller than that of the value before optimization, respectively. It is beneficial to save material and space of the reducers. Furthermore, the smaller of the width of cycloid gear, the less of the installation error.

4.3. Optimization based on maximum efficiency

A comparison study of the optimization based on maximum efficiency is also carried out in this work. The diameter of the cycloid gear center hole and width of the cycloid gear are constant values, which are chosen as 53.5 mm and 11 mm, respectively. Other design variables keep the same as Table 5. In the optimization process, the variation of efficiency is shown in Fig. 4. The best results for the optimization of the objective are listed in Table 7. From the obtained numerical results, the following conclusions can be made: (i) All of optimization variables can satisfy the constraint conditions. (ii) The value of the efficiency is increased by about 3.94%. (iii) Additional volume reduction of 7.88% is observed compared to the results before optimization. (iv) The value the

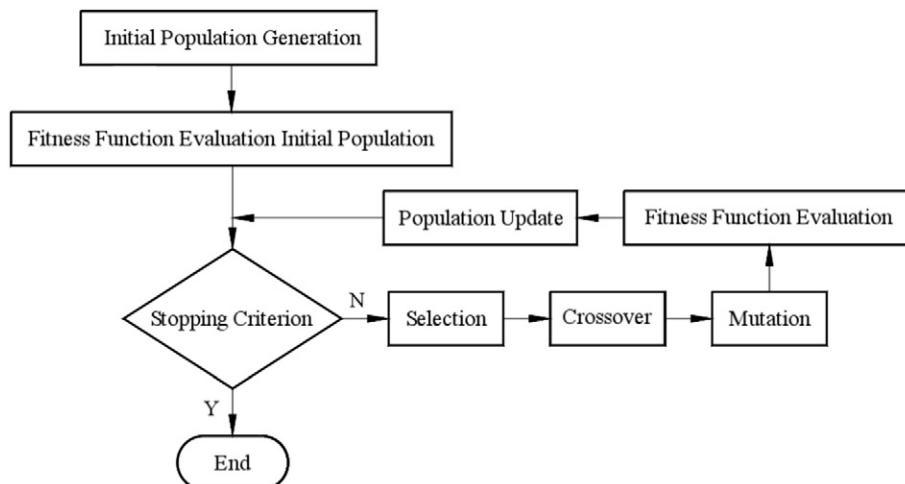


Fig. 2. Flowchart of Genetic Algorithms.

Table 5

Contrast gear parameters before and after optimization (Multi-objective optimization).

		Before optimization	After optimization	Rounded values
Design variables	Diameter of pin wheel central circle	144	140.1206	141
	Diameter of pin wheel	10	8.4743	8.5
	Width of cycloid gear	11	8.2116	8.3
	Diameter of cycloid gear center hole	53.5	54.6783	54.5
	Short width coefficient	0.611	0.8441	0.8441
	Diameter of pin central circle	90	99.9832	100
	Diameter of pin	12	13.1325	13.2
Constant value	Transmission ratio	43	43	43
Optimization objectives	Efficiency	0.862	0.8949	0.8947
	Volume	101,204	67,943	68,301

diameter of the pin wheel central circle is smaller than that of the value before optimization, resulting that the structure of the reducer is more compact.

5. Results analysis

In the optimization based on maximum efficiency, the width of cycloid gear, which is not a function of efficiency, should be given as a constant value. Although the obtained efficiency is the best in the above three optimizations, the axial size of the reducer is unrestricted, which will cause larger volume or heavy weight of reducers. Therefore, the optimization based on maximum efficiency is suitable for the occasions where high efficiency is required, and volume of reducer is unrestricted.

In the optimization based on minimum volume, the values the diameter of the pin wheel central circle and the width of the cycloid gear are smaller than that of the value before optimization, respectively. The results demonstrated that axial and radial sizes of the reducers are both reduced, resulting that the structure is more compact. In the optimization, the ranges of the width of the cycloid gear should be given. Otherwise, the cycloid gear attempts to turn thin in order to receive heavy load with small volume. This will certainly lead to bad stability and low efficiency. The optimization based on minimum volume is suitable for the occasions where small volume is required, and cost of the reducer is unrestricted.

The above two optimizations do not fare well over a broad spectrum of problem domains. Moreover, they are not robust. Due to the complex nature of optimization problem, these two optimizations are not ideal for solving these problems, as they tend to obtain a local optimal solution, while ignoring other performances of reducers. The multi-objective optimization refers to the solution of problems with two or more objective functions, which are normally in conflict with each other. As shown in Table 5–Table 7, comparing with the obtained results in the optimization based on maximum efficiency, the values of efficiency and volume obtained in multi-objective optimization are reduced by about 0.15% and 26.74%, respectively. On the contrary, the value of efficiency is 2.25% and the value of volume is 5.58% larger than that obtained in the optimization based on minimum volume. The multi-objective optimization results are controlled mainly by the balance between objectives. The results reveal that various aspects must be considered to obtain robust optimal solutions in multi-objective optimization.

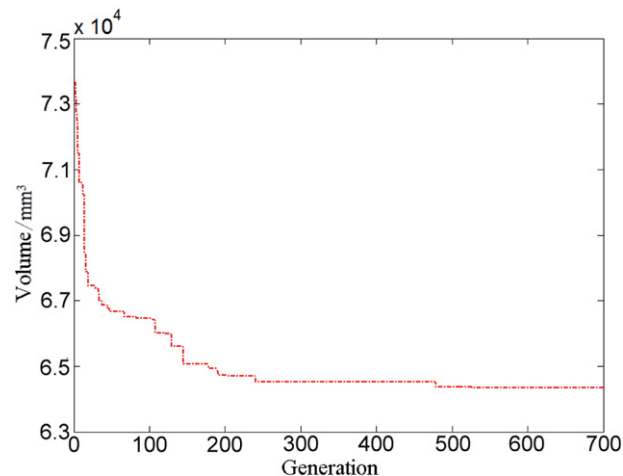
**Fig. 3.** Performance trace of optimal efficiency.

Table 6

Contrast gear parameters before and after optimization (Optimization based on minimum volume).

		Before optimization	After optimization	Rounded values
Design variables	Diameter of pin wheel central circle	144	140.0031	140
	Diameter of pin wheel	10	9.9081	9.9
	Width of cycloid gear	11	8.1432	8.2
	Diameter of cycloid gear center hole	53.5	54.862	55
	Short width coefficient	0.611	0.778	0.778
	Diameter of pin	12	13.2131	13.2
	Transmission ratio	43	43	43
Constant value	Diameter of pin central circle	90	90	90
Optimization objective	Volume	101,204	64,532	64,693
Other	Efficiency	0.862	0.876	0.875

6. Conclusion

The problem of minimum volume and maximum efficiency design of cycloid speed reducers has been a subject of considerable interest, since many high performance power transmission applications (e.g., industry robots, machine tools, automatic machinery, etc.) require small volume and high efficiency. An optimization methodology based on genetic algorithm is presented. The goal of the optimization is to simultaneously minimize volume and maximize efficiency. The mathematical model of multi-objective optimization, including the design variables, objective functions, and constraints is developed. The effectiveness of this optimization was demonstrated by an example. By applying the multi-objective optimization design of the cycloid speed reducer, a moderate increase in efficiency (3.79%) and a drastic reduction in volume (32.51%) are achieved. It is observed that the proposed multi-objective optimization design offer better design solutions.

As stated in Section 5, single optimizations are not ideal for solving the complex nature of optimization problems. They tend to obtain a local optimal solution, while ignoring other performances of reducers. Contrastingly, in multi-objective optimization, the optimization results are controlled mainly by the balance between objectives, which are normally in conflict with each other. In other words, various aspects must be considered to obtain robust optimal solutions. The obtained results reveal that it is more effective and feasible to use multi-objective optimization for designers to optimize complex systems such as reducers. Although gear reducers are used to derive the mathematical model of multi-objective optimization, in fact, the optimization objectives in the proposed methodology is not limited to reducers.

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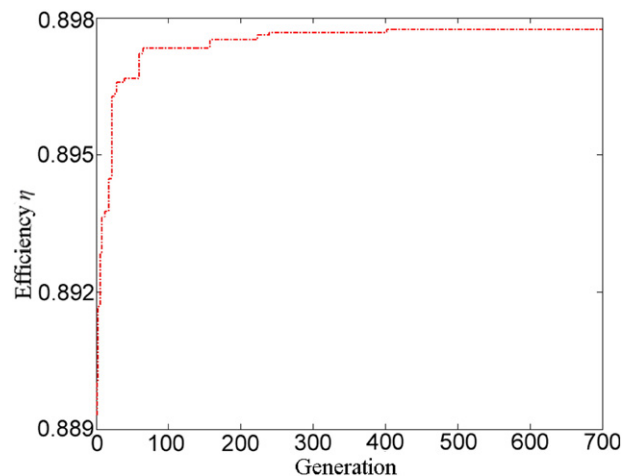
**Fig. 4.** Performance trace of optimal efficiency.

Table 7

Contrast gear parameters before and after optimization (optimization based on maximum efficiency).

		Before optimization	After optimization	Rounded values
Design variables	Diameter of pin wheel central circle	144	140.9316	141
	Diameter of pin wheel	10	7.6999	7.7
	Short width coefficient	0.611	0.8750	0.875
	Diameter of pin central circle	90	99.9567	100
	Diameter of pin	12	12.2048	12.2
	Transmission ratio	43	43	43
Constant value	Width of cycloid gear	11	11	11
	Diameter of cycloid gear center hole	53.5	53.5	53.5
	Efficiency	0.862	0.8971	0.896
Optimization objective	Efficiency	0.862	0.8971	0.896
Other	Volume	101,204	93,067	93,229

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